

本项目地址: https://github.com/soniccomm/sonic_application

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概述

本仓库提供了一个基于 MATLAB 的综合性超声成像处理工具箱, 专为处理平面波超声采集的原始数据而设计。它包含了四种先进的血流成像模式的后处理脚本, 用于从原始数据中生成丰富的血流动力学图像。

核心功能模块

- **UltrafastColorDoppler (超快彩色多普勒)**: 用于生成表征血流速度和方向的彩色血流图。
- **UltrafastPowerDoppler (超快能量多普勒)**: 用于生成表征血流存在和相对量的能量多普勒图像, 对低速血流更敏感。
- **UltrafastSpectralDoppler (超快频谱多普勒)**: 用于提取特定点的血流速度频谱, 进行定量分析。
- **UltrasoundLocalizationMicroscopy (超声定位显微技术/超分辨微血管成像)**: 通过追踪超声造影剂 (微泡) 实现超越衍射极限的超高分辨率微血管成像。

使用前准备: 数据采集

重要提示: 所有这些脚本都是后处理工具。运行它们之前, 您必须已经采集并保存了原始的平面波数据。

下载与设置

- 从 GitHub 下载整个项目仓库到您的本地计算机。

https://github.com/soniccomm/sonic_application

- 将 Application 文件夹置于 script 文件夹中, 将 lib 里面的文件移动到 script 中的 lib 文件夹
- 在 MATLAB 中, 将当前文件夹导航到您想要使用的特定脚本目录 (例如 .../Application/UltrafastPowerDoppler(能量多普勒)/)
- 打开脚本, 根据您的数据的参数 (数据路径等) 修改脚本开头的用户配置部分, 运行脚本

Overview

This repository provides a comprehensive MATLAB-based toolkit for ultrasound imaging processing, specifically designed to handle raw data acquired via **plane-wave ultrasound** transmission. It includes post-processing scripts for four advanced blood flow imaging modalities, enabling the generation of rich hemodynamic images from raw data.

Core Functional Modules

- **UltrafastColorDoppler**: Generates color flow maps depicting blood velocity and direction.
- **UltrafastPowerDoppler**: Produces power Doppler images highlighting the presence and relative density of blood flow, with enhanced sensitivity to low-velocity flow.
- **UltrafastSpectralDoppler**: Extracts velocity spectra at specific points for quantitative hemodynamic analysis.
- **UltrasoundLocalizationMicroscopy (ULM)**: Achieves super-resolution microvascular imaging beyond the diffraction limit by tracking ultrasound contrast agents (microbubbles).

Preparations: Data Acquisition

Critical Note: All scripts are **post-processing tools**. Users **must** have already **acquired and saved raw plane-wave data** before running these scripts.

Download & Setup

1. Download the Repository:

- Clone or download the entire project repository from GitHub
Link: https://github.com/soniccomm/sonic_application

2. Folder Structure Setup:

- Place the Application folder into the script directory.
- Move all files from the lib folder into the lib subfolder within the script directory.

3. **MATLAB Configuration:**

- Navigate MATLAB's *Current Folder* to the specific script directory you intend to use (e.g., .../Application/UltrafastPowerDoppler/).
- Open the target script and modify the user-defined parameters in the configuration section (e.g., data paths, acquisition settings) to match your dataset.
- Run the script to execute the processing pipeline.